Malitha Gunawardhana

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Summary

AI, Machine Learning and Data Professional with over four years of experience across machine learning, data science, analytics, operational reporting, and scalable technical systems. Skilled in Python, statistical modelling, deep learning, data validation, and translating complex datasets into practical insights and recommendations. Experienced in developing end-to-end AI and data pipelines, evaluating model performance, identifying root causes, and communicating findings to technical and non-technical stakeholders. Brings strong analytical thinking, technical depth, and multidisciplinary collaboration skills to help organisations improve decision-making, operational performance, and data-driven innovation.

Experience

Data Processing Technician, Part-Time — March 2026 – Present
Viotel Pty Ltd — *Parnell, Auckland*

- Developed machine learning-based anomaly detection workflows for accelerometer data collected from **8+ telecommunications towers**, supporting automated identification of abnormal structural vibrations.
- Trained and evaluated deep learning models for automated P- and S-wave phase picking using **3,729 seismic events and more than 30,000 station samples**, aiming to substantially reduce manual waveform-picking time.
- Collaborated with a three-member field team to install **17 seismic sensors** across the Taupo region, enabling successful deployment and continuous seismic data collection.

Machine Learning Engineer — Sep. 2022 – Jan. 2025
Institute of Fundamental Technological Research, Polish Academy of Sciences (IPPT PAN) — *Warsaw, Poland*

- Led the development and refinement of deep learning models for **tumour detection and segmentation**, working closely with clinical collaborators to improve model performance.
- Designed and implemented **scalable artificial intelligence pipelines for medical image analysis,** data processing, model training, and evaluation.
- Collaborated with more than **three cross-functional teams, including clinicians, researchers, and engineers**, to align artificial intelligence solutions with clinical and research requirements.

Artificial Intelligence Researcher — Sep. 2022 – Sep. 2023
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — *Abu Dhabi, UAE*

- Designed and implemented novel AI frameworks for perceptual understanding, contributing to the **development of robust self-supervised learning techniques**.
- Carried out advanced research in **network calibration**, **LLMs**, **video analysis**, **self-supervised learning**, and **semi-supervised learning**
- Led and collaborated with a four-member research team to produce high-quality research and contribute to publications in leading artificial intelligence conferences.

Machine Learning Engineer — June 2022 – Nov. 2022
PromiseQ GmbH — *Berlin, Germany*

- Improved the accuracy and reliability of a d**eep learning-based CCTV surveillance** system designed to detect activities in real time by integrating advanced object detection and anomaly recognition algorithms
- Reduced false alarms by **5%** using network calibration techniques, improving overall system reliability.
- Collaborated with cross-functional teams to **deploy scalable AI-driven surveillance solutions,** enhancing security and operational effectiveness

Full-Stack Software Engineer — March 2021 – May 2022
Xeptagon (Pvt) Ltd — *Colombo, Sri Lanka*

- Led team development of a **domain drop-catching system**, applying **optimized algorithm design** and **multi-threaded back-end engineering**, achieving over **90% success rate** and with less than **50ms latency**.
- Designed and implemented **scalable, responsive backend solutions** using an iterative feedback approach with stakeholders to ensure seamless system performance.
- Conducted feature extraction of 80+ audio signals, using **signal processing techniques** and feature engineering to power a student learning management system..

Software Engineer (Intern) — June 2019 – Dec. 2019
Synergen Technology Labs (Pvt) Ltd — *Colombo, Sri Lanka*

- Designed a wearable device for **physiological signal acquisition** and developed **feature extraction pipelines** to compute numerical stress values.
- Trained a **machine learning model** to classify stress into **relaxed, cognitive stress, physical stress, and emotional stress**, obtaining over **90%** stress categorisation accuracy.

Education

Doctor of Philosophy Dec. 2023 – Aug. 2026
University of Auckland *Auckland, New Zealand*

- Thesis: Robust, Uncertainty-Aware, and Scalable Deep Learning for Bi-Atrial Segmentation from LGE-MRI in Atrial Fibrillation
- **Award: Health Research Council Scholarship**
- Deep learning research to improve treatment outcomes of atrial fibrillation patients.
- Exploration of effective fusion mechanisms for pretrained and custom models
- Build efficient data pipelines for conducting ML experiments
- Publish and present research outcomes in internationally recognized journals and conferences

B.Sc. Engineering Honours Jan. 2017 – July 2021
University of Moratuwa *Moratuwa, Sri Lanka*

- Dean's list placement in Semester 7
- *Key Modules: Image Processing and Machine Vision, Neural Networks and Fuzzy Logic, Calculus, Linear Algebra, Differential Equations, Statistics, Graph Theory, Medical Imaging, Signal Processing.*

Certifications & Training

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional
  - ∗ Fundamentals of LLMs, model pre-training, fine-tuning and alignment, generative AI, and MLOps
- Microsoft Certified: Azure Fundamentals (AZ-900)
  - ∗ Azure cloud services, cloud computing concepts, storage, networking, security, and virtualisation
- Large Language Model Agents, UC Berkeley
  - ∗ LLM agents, autonomous AI systems, multi-agent coordination, prompt engineering, RAG, and MLOps
- Deep Learning Specialization, DeepLearning.AI
  - ∗ Neural networks, deep learning, convolutional neural networks, and recurrent neural networks
- Data Science Career Track, 365 Data Science
  - ∗ Python, SQL, data wrangling, statistical analysis, machine learning, and AI model evaluation

Technical Skills

Programming Languages: Python (Pandas, Numpy) , SQL, MATLAB, Go, JavaScript, Bash, R
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, YOLO, XGBoost
Generative AI and LLMs: LangChain, CrewAI, Weaviate, Retrieval-Augmented Generation, LLM agents
Data and Distributed Computing: PySpark, Apache Kafka, Apache Airflow
Data Analytics and BI: MLFlow, Power BI, Microsoft Excel
Developer and AI Tools: Docker, Google Cloud Platform, Git, Jira, Confluence, Linux, Claude, Cursor, Codex
Technical Domains: Deep Learning, Computer Vision, Medical Imaging, Video Analytics, Anomaly Detection, Time-series Analysis, Signal Processing, Predictive & Statistical Modelling, Model Deployment and Monitoring, Data Science & Analytics, MLOps, Vision-Language Models, Foundation Models

Service and Leadership

University of Auckland 2023 – Present

- Elected Student Council Member representing postgraduate students. Participated in university committee meetings, contributed to student welfare initiatives, and supported policy discussions.

IEEE Engineering in Medicine and Biology Student Branch Chapter at UoM 2018 – 2021

- Served as Advisor for ISC 2021 Moratuwa, Council Member, and Treasurer, supporting conference planning, chapter operations, and financial management.

Rotaract Club of UoM and Alumni of UoM 2016 – 2025

- Served as Vice President for Club Service and Club Service Director, led fellowship activities, received a Bronze award at the 2022 Rotaract District Assembly, and earned the Spirit of Service Award in 2017, 2018, and 2020.

References

Available upon request.